

00360015 - Finite Element Methods in Mechanical Engineering 1

Project - deformation of a mechanical membrane

Last Update: December 28, 2025

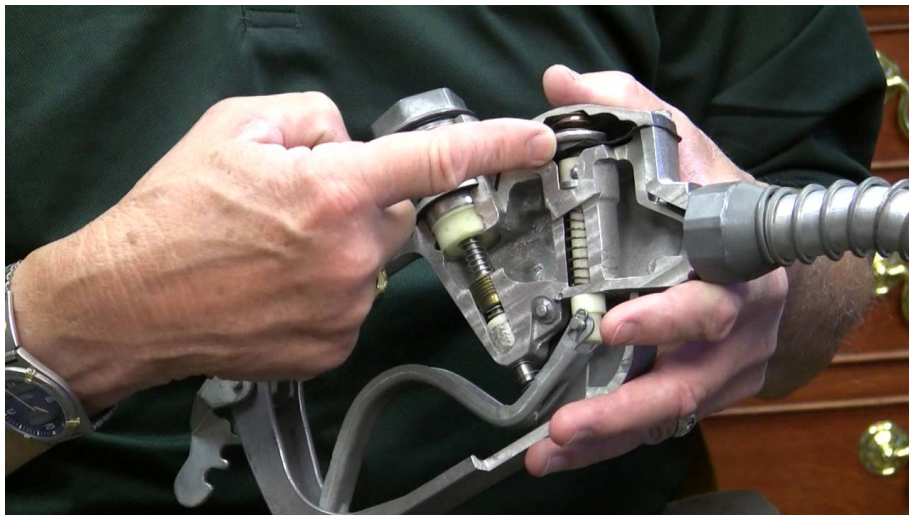


Figure 1: Example of a mechanical membrane in a gas fueling nozzle



Introduction

Before beginning the project, please read this document carefully to understand the required tasks and the structure of the final report.

The aim of this project is to use finite element tools to find the transverse displacement of a mechanical membrane on an elastic foundation. You will implement the full finite element workflow (pre-processing, processing, and post-processing) to solve the membrane equation on an arbitrary two-dimensional domain. This problem models a wide range of real systems: acoustic devices such as microphones and loudspeakers, fuel-pump diaphragms, and biological tissues resting on compliant substrates.

Project due date: 12 February 2026, 9:00 AM

- Submit the project codes and both a printed copy and an electronic version of your report.

Project defense: 16 February 2026, 9:00–12:00 AM

- A short written or oral examination in which you will answer questions about your report and demonstrate understanding of your implementation.



Mechanical Membranes

What is a mechanical membrane? A mechanical membrane is a structural element that its thickness t is very small as compared to in-plane characteristic size L ($t \ll L$), and can support no bending moments (similar to a one dimensional spring). All the loads are carried in-plane through tension stresses. If the membrane has uniform, isotropic pre-tension $T[N/m]$, and is loaded by a transverse pressure $p[N/m^2]$, the small-slope equilibrium reduces to a general class of Helmholtz-type equations:

$$\begin{cases} -T\nabla^2 w(x, y) + \kappa(x, y)w(x, y) = p(x, y) & \text{in } \Omega \\ w = \bar{w}(s) & \text{prescribed edge motion on } \partial\Omega \end{cases} \quad (1)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the Laplacian in the membrane plane, w is the transverse deflection of the membrane, and κ is the substrate stiffness underneath the membrane $[N/m^3]$ and s is a coordinate along the boundary. In **linear membranes**, tension is known and essentially constant (applies where there is dominant pre-tension).

Tip: If $p \approx p_0$ is roughly uniform and the domain has a characteristic size L ,

$$w_{typ} = O\left(\frac{p_0 L^2}{T}\right) \quad (2)$$



Structure of the Project

In this project you will solve [Equation 1](#) in a complex domain for which no analytical solution exists. The project will consist of your own matlab codes and an electronic submission of **typed** project report explaining the logic behind the numerous functions you will write. Each project report should include:

- **Cover page:** Project title and author name, ID, and email.
- **Introduction:** Describe the physical problem, and derive the finite element formulation of the problem as learned in class and using the class notations [1]. This includes writing the final weak formulation you will discretize; The structure of the elements used, the structure of their basis functions, and how the local coordinate system and node numbering are defined; Finally, provide the computational expressions for the element-level stiffness matrices ${}^{\kappa}K_e$ and mass matrices M_e as they are evaluated using Gauss quadrature.
- **Methods:** Explain how your own code works and how each major step contributes to solving the problem. Your description should be clear enough for someone to run and recover your results by running your project code. Where your code performs specific calculations, the mathematical formulation should be given in the text.

Additionally, show your code is capable of solving a simple problem similar to the one shown in class (lecture 10):

$$\begin{cases} -\nabla^2 u = f(x, y) & \text{in } \Omega = [0, 1] \times [0, 1] \\ u = 0 & \text{on } \partial\Omega \text{ (all around)} \end{cases} \quad (3)$$

where $f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$, and the exact solution is $u(x, y) = \sin(\pi x) \sin(\pi y)$.

- Using symmetry to model a quarter, with two elements, validate each computation step by comparing to an analytical solution.
- Mesh the quarter with 8, 32, 128, 512 elements and present the log-log graph of $\|e\|_0$ (L_2) and $\|e\|_1$ (H_1) norms of the solution.
- **Results**
- **Discussion and Conclusions**



Finite Elements Methods and Implementation

Benchmark problem

To validate your FE program produces correct results, it is good practice to first test it on a **benchmark problem** for which an analytical solution exists. In this project, we will use a problem of a circular membrane under uniform pressure:

$$\begin{cases} -\nabla^2 w + \kappa_0 w = p_0 & \text{in } \Omega \\ w = 0 & \text{on the fixed boundary } \partial\Omega \end{cases} \quad (4)$$

where κ_0 and p_0 are both constant values. For this case, the deflection is given by

$$w(r, \theta) = \frac{p_0}{\kappa_0} \left[1 - \frac{I_0(\sqrt{\kappa}r)}{I_0(\sqrt{\kappa}R)} \right] \quad (5)$$

where I_0 is a modified Bessel function of the first kind, R is the radius of the membrane, and r, θ are polar coordinates from the center of the circular membrane.

1 Pre-Processor

- Open "project geometries.pdf" file and find your personalized problem according to your I.D number.
- For both benchmark and project problems mesh the geometry using 9-node biquadratic Macro Elements, and **linear triangular elements** for the solution. Consider the symmetry of the geometries.
- For both benchmark and project problems mesh the geometry using 9-node biquadratic macro elements, and use **quadratic triangular elements** for the solution. Consider the symmetry of the geometries.

All the following computations of the entire project should be performed for each of the two element types.



- The input to the pre-processor is the geometry (and element type), material properties, and all loading conditions as defined in [Equation 1](#): $\kappa(x, y)$, $p(x, y)$.
- The output of the pre-processor is "Elements", "Nodes" and "Boundary" arrays. "Nodes" array contains global position of the nodes. "Elements" array contains global numbers of the nodes for each element. "Boundary" array contains node numbers, that are adjacent to the boundary and information about boundary type and value.
- Repeat both sections for both linear and quadratic elements.

2 Processor

- Use the output of the pre-processor to calculate the global matrices, force vector and apply the boundary conditions. Find the solution $\tilde{w} = w_i$ at the nodes.

3 Post-Processor

- Use the output of the processor to produce a figure for the deformation field of the mechanical membrane.

Tip: To create these figures, build a loop that calculates the solution $u_e = \mathbf{N}\mathbf{U}^e$, curvature and error in a local domain for each element separately, then plot the result in global domain using isoparametric mapping $x = \mathbf{N}\mathbf{X}$, $y = \mathbf{N}\mathbf{Y}$ (element by element).

- Present the log-log graph of $\|e\|_0$ (L_2) and $\|e\|_1$ (H_1) norms vs the characteristic size of an element h .
 - For the benchmark problem compare to the analytical solution.
 - For project problem use the best refinement result as a reference to calculate the norms.
- Show that the applied load from the pressure is in equilibrium with the total upward force provided by the elastic foundation

$$F_{load} = \int_{\Omega} p(x, y) d\Omega \approx F_{reaction} = \int_{\Omega} \kappa(x, y) w(x, y) d\Omega + \int_{\partial\Omega} q_n(s) ds \quad (6)$$



where $q_n(s)$ is the boundary reaction. Note that for every Dirichlet boundary node u_i , the corresponding entry $R_i = F_i - K_{ij}u_j$ is the reaction force at that node. Consequently,

$$\int_{\partial\Omega} q_n(s)ds = \sum R_i \quad (7)$$

- Clearly explain how you extract the reaction forces from the residual. Include units and physical interpretation.

4 Eigenvalue Problem

Consider a rectangular membrane (fixed at the edges) as a benchmark problem. Find the first four eigenvalues and plot the corresponding eigenmodes $\lambda_i^h, i = 1, 2, 3, 4$ using the finite element approach, compare with analytically obtained eigenvalues [2] and present a convergence graph on a log-log plot. Do the same for both linear triangular and quadratic triangular element types.

5 Discussion and Conclusions

In this section you will interpret the numerical results obtained throughout the project and summarize the main findings. Critically discuss the results you obtained. Were there any problems you encountered with these results? Discuss the accuracy and convergence behavior of your finite element solutions, and relate the observed trends to the expected theoretical performance of the element types used.

Conclude by evaluating the strengths and limitations of your implementation and briefly suggesting possible improvements or extensions.

Tip: Thoughtful additions, such as deeper insights into the finite element process, comparisons with theory, or further research into identified effects will not go unnoticed and can contribute positively to your evaluation.

Good luck!



Bibliography

1. All lecture notes and tutorial materials from the course.
2. Russell, [Vibrational Modeshapes of a Rectangular Membrane \(fixed at the edges\)](#). Acoustics and Vibration Animations, 2018